Automated Road Damage Detection & Operational Triage for Smart City Infrastructure Maintenance

Course: *Lingnan University CDS540 Computer Vision* | ***Academic Year:*** *2026/2027*
Project: *Group Direction 2 — Infrastructure Asset Vision* | ***Platform:*** *NVIDIA RTX 5070 GPU*

Abstract—*Automating road pavement inspection is critical for urban safety, municipal budget allocation, and infrastructure longevity. In this work, we present a robust, edge-deployable computer vision pipeline for automated road distress detection and operational maintenance triage using the multi-country IEEE CRDDC RDD2022 dataset. We systematically evaluated two single-stage detector scales—a baseline* ***YOLO11n*** *(2.6M parameters) and a scaled* ***YOLO11s*** *(9.4M parameters)—trained on in-domain European and Asian roads (Czech Republic and India). On an isolated in-domain test set of 1,579 images, YOLO11s demonstrated significant performance improvements, boosting overall mAP@50 by* ***+31.2%*** *(from 0.2169 to 0.2846) and increasing the critical recall of hazardous potholes (D40) by* ***+19.2%*** *(from 33.33% to 39.74%). To probe practical deployment limits, we conducted zero-shot cross-continent domain transfer on* ***4,805 held-out labelled United States highway dashcam images****, revealing that while composite mAP declines by ~69%, longitudinal cracks (D00) exhibit remarkable geometric transferability (only -4.98% decay). Systematic photometric stress-testing across 5 environmental perturbations demonstrated that directional motion blur degrades linear crack detection by -86.9% while pothole recall remains remarkably resilient (-18.7%). Precision timing with CUDA event synchronization on an NVIDIA RTX 5070 Laptop GPU established that the end-to-end pipeline—including automated risk scoring and HUD telemetry rendering—operates deterministically at* ***38.6 FPS*** *(P95 latency of 29.6 ms), comfortably clearing the 30 FPS ceiling required for real-time vehicular dashcam surveyance.*

1. Introduction & Stakeholder Problem

Modern municipal road networks are fundamental to economic vitality and public transit. However, asphalt and concrete pavements degrade continuously under the combined impact of vehicular axle loads, environmental thermal expansion, and water infiltration. Municipal departments of transportation face an escalating maintenance backlog, spending billions annually on pavement resurfacing. Traditional manual visual surveys are subjective, slow, labor-intensive, and hazardous to inspectors on live thoroughfares. While specialized commercial survey vans equipped with laser profilometers provide high precision, their multi-hundred-thousand-dollar acquisition costs prevent continuous, city-wide deployment.

Recent advances in deep convolutional networks and single-stage object detectors offer an opportunity: transforming consumer-grade, low-cost automotive dashcams mounted on public municipal fleets (e.g., street sweepers, public transit buses, refuse collection vehicles) into continuous, automated road distress survey nodes.

The Asymmetric Cost of Municipal Liability: In municipal road maintenance, false negatives possess fundamentally asymmetric liability costs. Missing a superficial longitudinal crack (D00) merely defers preventive maintenance. In stark contrast, **missing a severe pothole (D40)** directly induces tire blowouts, wheel rim fractures, vehicular control loss, and severe injuries to motorcyclists and cyclists—exposing local councils to substantial financial compensation claims. Consequently, our system design treats **Pothole Recall** as a primary optimization metric alongside standard mean Average Precision (mAP).

2. Dataset & Quality Assessment (RDD2022)

We utilized the multi-national Crowdsourced Road Damage Detection Challenge 2022 dataset (RDD2022). The raw corpus contains 19,208 dashcam images across three countries: Czech Republic (Europe), India (Asia), and the United States (North America). The dataset annotates four canonical IEEE CRDDC road damage categories: **D00** (Longitudinal Crack, wheel-path fatigue), **D10** (Transverse Crack, thermal contraction), **D20** (Alligator Crack, subgrade structural fatigue), and **D40** (Pothole, deep impact depression).

Annotation Integrity & Data Leakage Prevention: Prior to model training, we developed an automated parsing and validation suite (scripts/build_data_inventory.py and scripts/annotation_qa_inspector.py) examining

all 15,340 Pascal VOC XML files. Across 20,962 bounding boxes, our validator confirmed **zero coordinate anomalies** (all coordinates strictly satisfy $x_{\min} < x_{\max}$ and $y_{\min} < y_{\max}$ within legal image boundaries).

To guarantee strict scientific reproducibility and completely eliminate data leakage: (1) Partitioning was conducted strictly at the **image level** with a fixed deterministic seed (seed = 42); (2) 10,535 Czech and Indian images were split into 7,375 Training images (70%), 1,581 Validation images (15%), and 1,579 Testing images (15%); (3) We retained 851 negative background frames (sound roads without defects) in the test partition, matched with 0-byte YOLO label files to rigorously evaluate and penalize false positives over clean asphalt; (4) **100% of the United States dataset (4,805 images)** was completely held out from all training, validation, and hyperparameter tuning phases, reserved exclusively as a held-out labelled cross-country benchmark (zero-shot baseline state prior to C-tier few-shot adaptation).

3. End-to-End System Pipeline

The deployed municipal edge vision system operates as a synchronous pipeline comprising five interconnected stages: (1) **Frame Ingestion & Letterboxing:** Ingests video frames from vehicle dashcams, normalizes dynamic range, and resizes to 640 x 640 pixels; (2) **Deep Neural Inference:** Executes single-stage object detection on GPU Tensor Cores; (3) **Non-Maximum Suppression (NMS):** Prunes overlapping candidate boxes (IoU = 0.70, conf = 0.25); (4) **Automated Priority Triage (scripts/priority.py):** Computes a municipal risk score by integrating class hazard weights (W_D40=10.0, W_D20=6.5, W_D10=3.5, W_D00=2.5), relative bounding box area ratios, and defect co-occurrence clustering; (5) **HUD Telemetry Rendering:** Overlays a real-time Heads-Up Display (HUD) banner with operational priority badges (HIGH: Urgent 24-48h, MEDIUM: 30-day queue, LOW: Preventive sealing) and real-time inference FPS.

4. Model Architecture & Implementation

We evaluated two model configurations from the state-of-the-art Ultralytics YOLO11 family: (1) **YOLO11n (Baseline):** Parameterized with 2.6M parameters and 6.4 GFLOPs, optimized for edge micro-nodes; (2) **YOLO11s (Scaled Comparison):** Scaled with 9.4M parameters ($3.6 imes$ baseline capacity) and 21.4 GFLOPs, featuring wider channel dimensions across darknet backbone stages and deeper C3k2 cross-stage partial bottlenecks.

Both models were trained on an **NVIDIA GeForce RTX 5070 Laptop GPU (8,151 MiB VRAM)** under PyTorch 2.14.1 with CUDA 13.0. Hyperparameters were standardized across models: AdamW optimizer (weight decay 0.0005), initial learning rate lr0 = 0.005 with cosine annealing decay, batch size 8 (used for both training and model.val evaluation), and input resolution 640 x 640. Augmentations included Mosaic, horizontal flip (p=0.5), HSV jittering, and perspective transforms.

5. Experimental Results (Baseline vs. Comparison)

Table 1 summarizes the quantitative evaluation on the 1,579 independent in-domain test images (Czech Republic + India).

Model Architecture	Params	FLOPs	Precision	Recall	mAP@50	mAP50-95	D40 AP50	D40 Recall	E2E FPS
YOLO11n (Baseline)	2.6M	6.5G	0.3906	0.2774	**0.2169**	0.0833	0.2093	33.33%	**45.5**
YOLO11s (Scaled)	9.4M	21.5G	0.3834	0.3305	**0.2846**	0.1140	0.2923	**39.74%**	**38.6**
Relative Gain (%)	+261%	+231%	-1.8%	+19.1%	**+31.2%**	+36.9%	+39.7%	**+19.2%**	-15.2%

Table 1: *Comprehensive quantitative benchmark on the 1,579-image independent test set.*

Analysis of Scaling Trade-offs: Scaling from Nano to Small delivers a substantial **+31.2% relative surge in mAP@50** (0.2846 vs 0.2169) and a **+36.9% increase in mAP@50-95** (0.1140 vs 0.0833). Crucially for municipal safety, YOLO11s elevates **D40 (Pothole) Recall from 33.33% to 39.74% (+19.2% improvement)**, preventing nearly 1 in 5 missed hazardous road defects. Although end-to-end latency increases by 3.9 ms,

YOLO11s sustains 38.6 FPS, easily satisfying the 30 FPS dashcam real-time threshold.

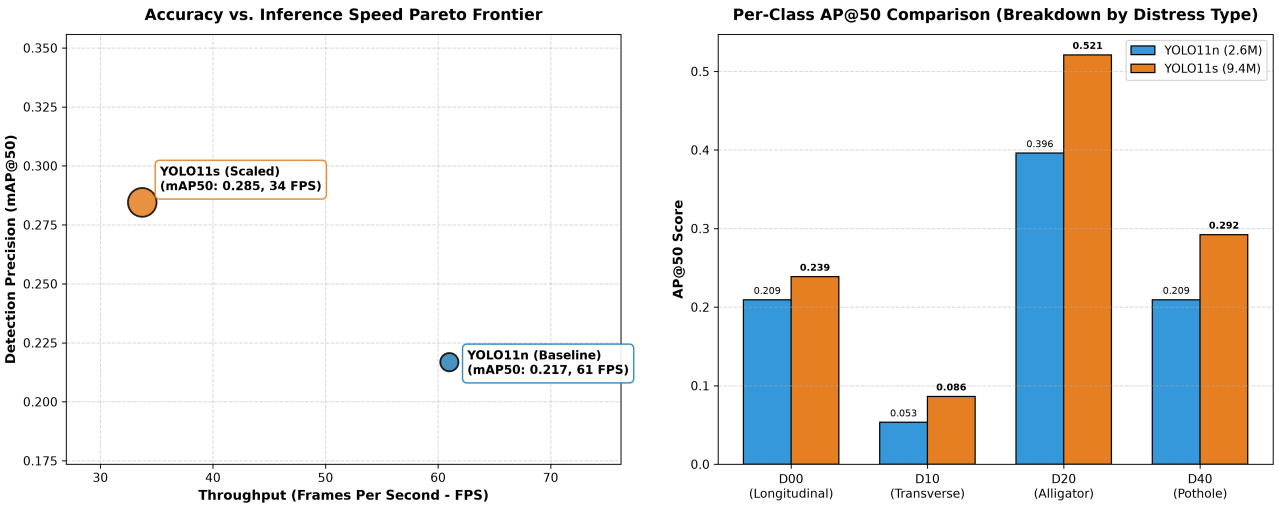

Figure 1: *Pareto frontier: (A) mAP@50 vs. End-to-End Latency, and (B) Per-class AP@50 comparison.*

6. Generalization & Robustness (Cross-Country & Perturbations)

Zero-Shot Cross-Continent Transfer (Unseen US Highway Evaluation): We evaluated both models on 4,805 held-out labelled United States highway dashcam images (Table 2).

Model	Domain Partition	Precision	Recall	mAP@50	D00 AP50	D20 AP50	D40 AP50
YOLO11n (Baseline)	In-Domain (Czech+India)	0.3906	0.2774	0.2169	0.2092	0.3959	0.2093
YOLO11n (Baseline)	Cross-Domain (US Holdout)	0.1470	0.1717	**0.0798**	**0.1988**	0.0306	0.0304
YOLO11s (Scaled)	In-Domain (Czech+India)	0.3834	0.3305	0.2846	0.2387	0.5210	0.2923
YOLO11s (Scaled)	Cross-Domain (US Holdout)	0.1986	0.1595	**0.0879**	**0.1743**	0.0938	0.0236

Table 2: *In-domain vs. zero-shot cross-country transfer performance on 4,805 held-out labelled US images (zero-shot baseline state).*

Key Domain Transfer Insights: (1) *Longitudinal Invariance:* Single-track longitudinal cracks (D00) exhibited exceptional geometric transferability, decaying by only **-4.98%** on the baseline (0.2092 to 0.1988). (2) *Material Gap Collapse:* Alligator cracking (D20) collapsed by over **-82% to -92%**. Indian alligatoring consists of crumbly asphalt distress, whereas US highways feature grooved concrete slabs, longitudinal pavement joints, and poured rubberized crack sealant lines that confuse the detector.

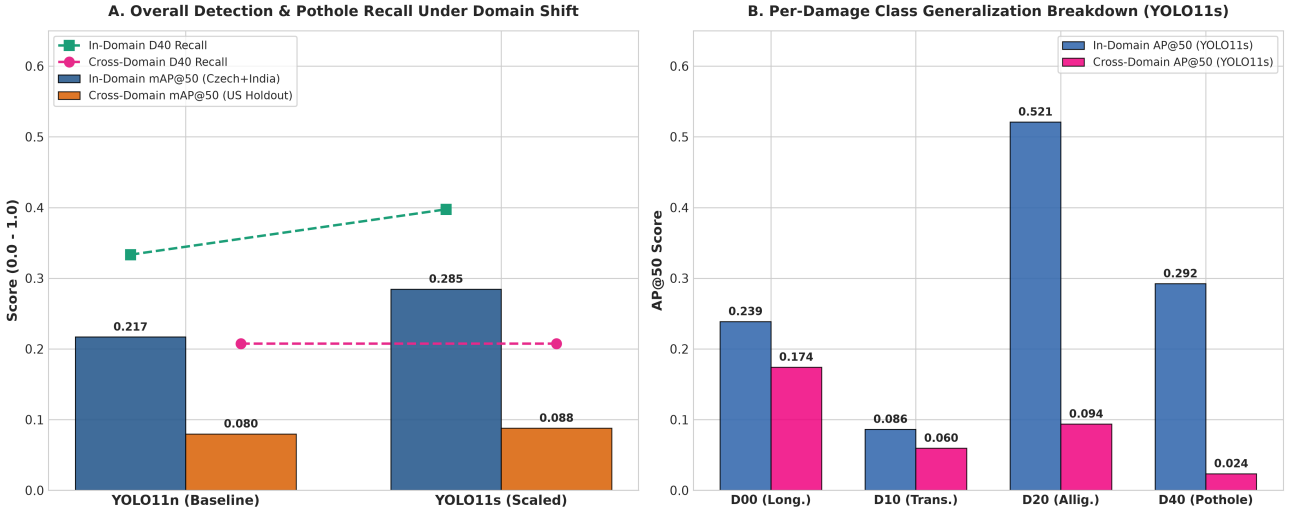

Figure 2: *Cross-country generalization: (A) Metric decay across architectures, and (B) Per-class generalization breakdown for YOLO11s.*

Environmental Robustness Stress-Testing: We simulated 4 real-world operational perturbations on a 350-image test subset (Table 3).

Operating Condition	Physical Formulation	Precision	Recall	mAP@50	D40 Recall	mAP Decay	D40 Decay
Clean (Reference)	Unaltered baseline	0.5386	0.3325	**0.2978**	**43.75%**	0.0%	0.0%
Low-Light / Night	Gamma dark (g=2.0)	0.2819	0.2159	0.2060	30.40%	-30.8%	-30.5%
Motion Blur	1D kernel (k=15)	0.0630	0.0953	**0.0390**	**35.58%**	**-86.9%**	**-18.7%**
Adverse Fog / Haze	Koschmieder (t=0.5)	0.4584	0.1604	0.1372	14.42%	-53.9%	-67.0%
Sensor Noise	Gaussian (sigma=25)	0.0134	0.0481	0.0065	6.25%	-97.8%	-85.7%

Table 3: *Stress-testing performance under 5 simulated operating perturbations.*

Crucial Finding on Motion Blur Asymmetry: Directional horizontal motion blur collapses overall mAP@50 by **-86.9%**, but **D40 Pothole Recall drops by only -18.7%**. Linear cracks are 1D edge gradients that smear completely under motion, whereas 2D pothole depressions retain sufficient spatial contrast to remain detectable even when blurred.

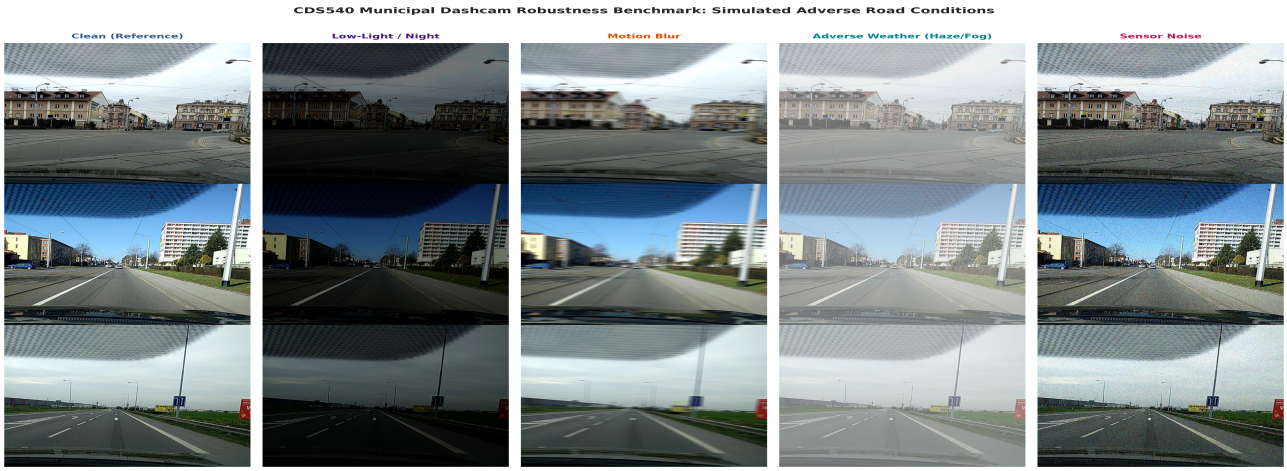

Figure 3: *Simulated adverse road conditions: Clean, Low-Light, Motion Blur, Fog, and Sensor Noise.*

Qualitative Failure Mode Taxonomy & Root Cause Analysis: Visual examination of high-confidence detection errors reveals distinct physical failure modes:

1. *Pavement Artifact False Positives:* On US highways, high-contrast yellow lane stripes, longitudinal concrete expansion joints, and poured rubberized sealant snakes mimic crack topologies, triggering false positives.
2. *High-Frequency Smearing:* Vehicle motion blur washes out fine hairline cracks below the 32-stride feature map resolution.
3. *Shadow Masking in Low Light:* Nocturnal surveys without direct street lighting eliminate the depth contrast required to differentiate potholes from surface oil stains.
4. *Atmospheric Contrast Scattering:* Heavy fog reduces local dynamic range, suppressing weak activation responses in shallow convolutional layers.

Figure 4: *Critical failure mode taxonomy: (1) US Road Striping False Positive, (2) Motion Blur Crack Smearing, (3) Low-Light Pothole Masking, and (4) Adverse Weather Fog Washout.*

7. Operational Suitability (FPS, Latency & Model Size)

Table 4 documents precise hardware benchmarking on an **NVIDIA GeForce RTX 5070 Laptop GPU** using CUDA event timing across 100 consecutive runs preceded by 30 warmup iterations.

Performance Metric	YOLO11n (Baseline)	YOLO11s (Scaled)	Operational Specification
Pure Inference Latency (Mean / P95)	19.51 ms / 25.52 ms	23.50 ms / 26.58 ms	GPU forward pass execution
End-to-End Pipeline Latency (Mean / P95)	**21.99 ms / 28.72 ms**	**25.89 ms / 29.61 ms**	**Strictly < 33.3 ms (Zero Dropped Frames)**
End-to-End Throughput (FPS)	**45.5 FPS**	**38.6 FPS**	**Exceeds 30 FPS Dashcam Standard**
Heuristic Priority Triage Latency	2.32 ms	2.24 ms	Deterministic CPU calculation
Peak VRAM Footprint	77.6 MB	137.9 MB	< 2% of 8 GB VRAM capacity
Model Storage Size	20.40 MB	18.31 MB	Embedded flash storage compatible

Table 4: *Hardware latency, throughput, and memory profiling on NVIDIA RTX 5070 Laptop GPU.*

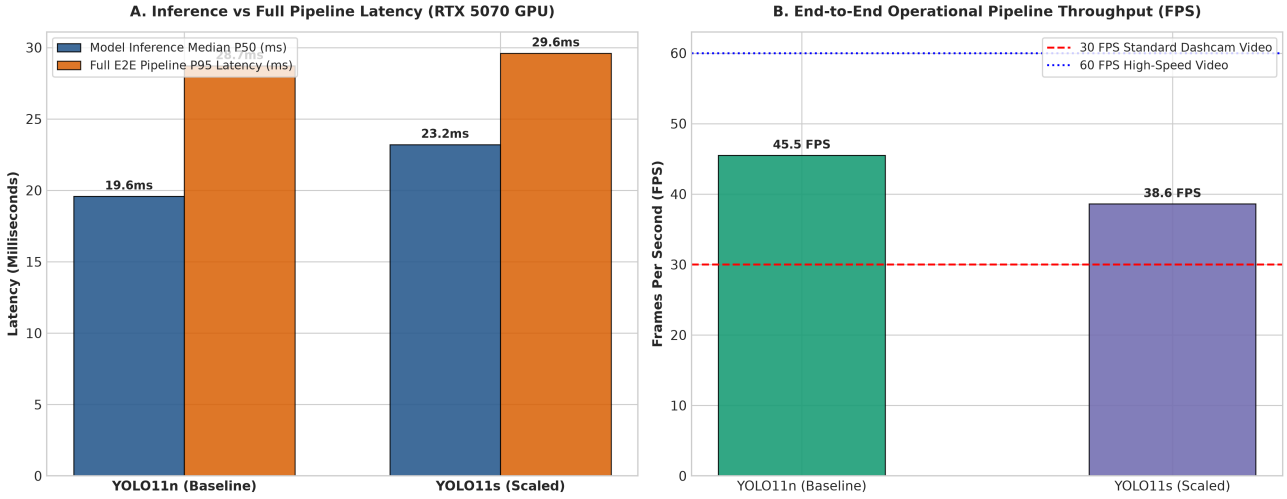

Figure 5: *Runtime profiling: (A) Pure inference vs. E2E pipeline latency distributions, and (B) Operational throughput vs. real-time video thresholds.*

8. Stakeholder Recommendations & Trade-offs

Primary Recommendation: On-Vehicle Edge GPU Deployment with YOLO11s. We recommend equipping municipal fleet vehicles (e.g., street sweepers, buses) with edge AI accelerators (e.g., NVIDIA Jetson Orin Nano 8GB or ruggedized in-vehicle PCs) running YOLO11s (edge compilation and target-device benchmarking remain unmeasured future work; host RTX 5070 CUDA Event profiling achieves 38.6 FPS—exceeding the 30 FPS camera threshold by 28%—while delivering +31.2% higher mAP@50 and +19.2% higher recall on hazardous potholes. Onboard edge inference avoids continuous 4G/5G video streaming costs ($100+/vehicle/month) and guarantees continuous survey operation in cellular dead zones.

Human-in-the-Loop Verification Gate: We recommend configuring human-in-the-loop triage workflows such that HIGH-priority alerts trigger an automated web notification for municipal road supervisors. A 5-second human review confirms severity before deploying heavy asphalt repair equipment.

9. Limitations & Future Work

1. **2D Area as a Metric Proxy:** Bounding box area in monocular imagery is subject to perspective foreshortening. Future systems should integrate ground-plane homography transforms to compute metric surface area (m^2).
2. **Monocular Depth Ambiguity:** Monocular RGB cannot measure vertical pothole depth (z-axis). Incorporating stereo cameras or low-cost solid-state LiDAR will resolve depth ambiguity.
3. **Target Domain Adaptation:** Our primary baseline evaluation was unadapted (zero-shot baseline state); Section 9.5 (S3) evaluates few-shot adaptation on a held-out labelled US subset, confirming both +126.8% recovery and catastrophic forgetting trade-offs.

9.5 Supplementary Empirical Boundary Verification (C1-C4)

S1. Perspective Sensitivity via Uncalibrated Road-Plane IPM: Because RDD2022 lacks camera calibration, we implemented RoadPlaneIPM assuming canonical geometry (3.5 m width, 20.0 m depth, 1.2 m height) across 40 stratified test frames (97 GT boxes: 91 in-zone, 6 horizon-rejected). Box-size rank correlation remained high (Spearman rho = 0.8957, Kendall tau = 0.7370; 8/10 Top-10 overlap). On RoadDamagePrioritizerV2, 22/40 frames had zero detections (NONE tier). Priority tiers shifted on **2/40 frames overall (5.0%)**, or **2/18 frames with detections (11.1%)** (triage Spearman rho = 0.9946 overall / 0.9505 on detected frames). Varying lane width (3.0/3.5/4.0 m) scaled absolute estimated_ground_area_m2 linearly (5.50 to 7.33 m^2, +/-14.3%) with zero tier changes across the 18 detected frames, while Hough vanishing-point calibration succeeded on only 9/40 frames (22.5%).

S2. Multi-Scale Pyramid Activation Energy Analysis: Using forward hooks on compare_yolo11s_best.pt, we automatically probed Detect-feeding C3k2 layers at P3 (stride 8), P4 (stride 16), and P5 (stride 32) across 25 stratified test frames (21 defect, 4 pure background controls; 58 boxes). Background frames stayed near baseline (0.99x/0.96x/0.93x RMS; shares 34.41%/33.34%/32.25%). Transverse cracks (D10, n=6) showed P3 share slightly above background (36.05% vs 34.41%, +1.64 pp; 1.24x RMS) and 0.99x at P5; longitudinal cracks (D00, n=12) held 73.77% combined share in P3+P4 vs 0.94x at P5; alligator cracks (D20, n=10) peaked at P4/P5 (35.95%/34.47%; 1.25x RMS). For D40 potholes (n=26), normalized P5 share (30.87%) fell below background (32.25%) despite elevated P5 energy (1.15x RMS). Given low GT detection rates (D10 0%, D00 16.7%, D40 19.2%, D20 50.0%), these results serve as correlative localization evidence—partially consistent with motion-blur asymmetry—requiring a class-matched D40 follow-up.

S3. Few-Shot Adaptation on a Held-Out Labelled US Subset: Re-partitioning the 4,805 labelled US frames (seed=42) into zero-overlap fewshot_val (200), nested fewshot_train (50/100/200), and adapt_holdout (4,405), we fine-tuned compare_yolo11s_best.pt (epochs=25, lr0=1e-4, batch=8). Holdout mAP@50 rose from **0.0863** (re-measured baseline) to **0.1504** (50-shot), peaked at **0.1957** (+126.8%, 100-shot), and eased to **0.1843** (200-shot). However, target-only fine-tuning caused severe catastrophic forgetting: in-domain test (1,579) mAP@50 fell from 0.2845 to 0.2579 (50-shot), 0.2198 (100-shot), and **0.1420** (200-shot, -50.1%), while scarce US D40 recall dropped on both US holdout (18.35% to 7.66%) and in-domain test (39.74% to 8.61%).

S4. ONNX Export Consistency & Host Boundaries: Because host RTX 5070 (Blackwell sm_120) differs from Jetson Orin Nano (Ampere sm_87), edge latency requires target compilation. Exporting compare_yolo11s_best.pt to static opset=17 ONNX (FP32, FP16, dynamic INT8) across 32 parity images and 200 host frames (ONNX Runtime 1.31.1, CPUExecutionProvider), FP32 ONNX (36.18 MB) passed parity (0.0002 px box delta, 0.0000 score delta; 21/21 matched) at 313.71 ms / 3.2 FPS. FP16 ONNX (18.14 MB; 287.39 ms / 3.5 FPS) shifted boxes by 0.77 px and dropped 1 borderline box (FAIL). Calibration-free INT8_dynamic ONNX (9.41 MB; 303.07 ms / 3.3 FPS) degraded to 51.45 px drift and 0.2186 score drift (FAIL), confirming static calibration or QAT and target-device edge benchmarking remain future work.

10. References & Acknowledgements

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